



East West University

Department of Computer Science & Engineering

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Lab Assignment : Stack & Queue

Course Code : CSE207

Course Title : Data Structures

Instructor : Md. Manowarul Islam, Adjunct Faculty, Department of CSE

Objective:

The objective of this lab is to provide a fundamental idea about the Stack and Queue using array in C programming. At the end of the lab, students are able to:

- Implement push, pop, peek, display operations of Stack.
- Implement enqueue, dequeue, peek, display Operations of Queue.

Lab Tasks

1. Write a program in C to implement a stack that only allows even numbers to be pushed. Odd numbers should be rejected with an appropriate message.

Sample Input	Sample Output
Push 20	Stack Contain: 20
Push 23	Cannot be Inserted
Push 40	Stack Contain: 20 40

2. Write a program in C to Create a stack with a constraint that the **sum of all elements** in the stack should never exceed **100**. Prevent further pushes if adding the new element exceeds this limit.

Sample Input & Output
<i>If $current_sum + new_element > 100$, reject the push and display: "Sum limit exceeded."</i>

3. Write a program in C to Push N elements to the stack, then implement a function to reverse the **top K elements** without using any extra stack or array.

Sample Input	Sample Output
Push : 10 20 30 40 50 K=3	Reversed Stack : 10 20 50 40 30

4. Write a program in C that implements a stack which **ignores duplicate values** during push. If an element already exists in the stack, it should not be added again.

Sample Input	Sample Output
Push 10	Element 10 pushed to stack.
Push 20	Element 20 pushed to stack.
Push 30	Element 30 pushed to stack.
Push 20	Duplicate entry ignored.
Push 40	Element 40 pushed to stack.
Push 10	Duplicate entry ignored.
Display()	Stack Contain : 10 20 30 40

5. Write a program in C where in addition to normal peek (top), you can also **peek the middle element** of the stack.

Sample Input	Sample Output
Push 10	Element 10 pushed to stack.
Push 20	Element 20 pushed to stack.
Push 30	Element 30 pushed to stack.
Display()	Stack Contain : 10 20 30
Special_peek()	Middle element is 20.

6. Write a program in C to implement a queue that only allows prime numbers to be enqueued. Non-prime numbers should be rejected with an appropriate message.

Sample Input	Sample Output
Enqueue 7	Element 7 inserted to Queue
Enqueue 10	10 is not a prime Number
Enqueue 13	Element 13 inserted to Queue.
Display()	Queue Contains: 7 13

7. Write a program in C to implement a queue where the sum of all elements should never exceed 50. Reject new elements that would cause the sum to exceed 50.

Sample Input	Sample Output
Enqueue 20	Element 20 inserted to Queue
Enqueue 15	Element 15 inserted to Queue.
Enqueue 18	Inserting 18 would exceed sum limit (current sum = 35). Rejected
Enqueue 10	Element 10 inserted to Queue.
Display()	Queue Contains: 20 15 10